

End-to-End Wireless Multimedia Transmission System

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Abstract—Multimedia transmission over wireless networks is a hot research field these years, but little work focuses on discussion of efficient transmission between wireless terminals. Limited bandwidth of wireless networks and limited capabilities of wireless terminals are two major problems for the efficient transmission of multimedia flows. In this paper, we propose a system, End-to-End Wireless Multimedia Transmission (EEWMT) system, that supports end-to-end wireless transmission for multimedia flows and provides services to resolve the above two problems. EEWMT system is mainly made up of three parts, *anyDevice*, *Center Server* and *Agent Server*: *anyDevice* is a protocol stack working on terminals; *Center Server* provides indexing and database services and is responsible for session management; *Agent Server* provides data buffering and multimedia transcoding services. Implementation of our system shows that it enhances both the communication efficiency and the content quality

Keyword—wireless multimedia transmission; end-to-end system

I. INTRODUCTION

In recent years, wireless technologies are growing rapidly. Not only new wireless networks are built such as Wireless LAN and the third generation wireless networks (3G [1]), but also more and more powerful wireless terminals are developed such as camera mobile phones and Personal Digital Assistants (PDA). With the improvement of wireless technologies, more people begin to enjoy wireless applications and services. However, applications and services on wireless networks are still limited and not as attractive as those on Internet. Many researchers and developers are thinking one question, that is, how to make the wireless networks as popular as Internet.

One reason of the popularity of Internet is that people could obtain most amount of information from it. Nowadays, millions of people are enjoying searching songs, movies, games and other kinds of multimedia resources and sharing/exchanging them with others. Besides, communication with friends and strangers through Internet is also very attractive and useful to many people. Emails, instant messages and Bulletin Board System (BBS) etc., all these applications and services have even been parts of many people's daily life.

Entertainment applications such as online movies, living TV, network radio and audiovisual conversation are catching more and more people's hearts. However, up to now it is still hard to accomplish most of these over wireless networks. Wireless applications and services are most designed for people to access limited web resources in Internet. Little work is done to enable multimedia communication, especially end-to-end multimedia transmission. Compared with wired Internet, there are several obstacles that make end-to-end wireless multimedia transmission difficult to achieve:

1) *Low capability and limited resource of terminals*

Although with the development of WiFi more and more personal computers can join the wireless networks, most of wireless terminals are still PDAs and mobile phones. These wireless terminals typically have a small liquid crystal display supporting quite few colors and low resolution, a slow processor and a very small size of memory space etc. However, multimedia applications usually require high graphic processing capability, large size of memory space and also a big screen to display pictures and videos. Obviously, wireless terminals can only support limited multimedia applications.

2) *Diversity of wireless terminals*

In the Internet, although there are many types of multimedia data that are in different data formats, computers are still able to handle most of multimedia data formats. However, wireless terminals often support only a limited set of data formats due to their low capability and limited resource. The main reason is that manufacturers of wireless devices often focus on their portability and mobility, and their multimedia processing capabilities are sacrificed. So that when two different wireless terminals communicate with each other, their supported data formats may be not compatible and the communication cannot succeed. Even if some data from one terminal could be converted into the format supported by the receiver, such conversion often costs much and is not practical or acceptable to resource-limited wireless terminals.

3) *Low bandwidth of wireless networks*

As today's three most popular wireless networks, GPRS can provide a dedicated bandwidth to each subscriber up to 56Kbps, Wireless LAN can provide a shared bandwidth up to 11Mbps which may be quite lower when many people use it simultaneously, and 3G provides an adaptive bandwidth which can rise to 2Mbps when used home and falls to 128Kbps if the

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user is moving. Note that wireless devices are usually of low capabilities, multimedia streams created by wireless devices can not be compressed much and need high bit rate for real-time transmission. Therefore, it is really difficult to develop some real-time multimedia communication applications such as audiovisual conversation and videoconferencing.

4) *Fluctuated Bandwidth and Blackouts of wireless connections*

Multimedia data transmission, especially real-time transmission, requires steady high bit rate and is intolerant of package delay. On the contrary, wireless networks have fluctuated bandwidth and high probability of traffic congestion. Usually, the transmission time of multimedia data in a session is quite long, but the blackout of wireless connection may cause frequent reconnection and data retransmission.

In the design of EEWMT system, the above problems are well considered. The three main parts of EEWMT system are *anyDevice*, *Center Server* and *Agent Server*: *anyDevice* is a protocol stack working on terminals and it helps terminals to communicate with servers; *Center Server* provides indexing and database services and its session management module can help terminals quickly restart data transmission; *Agent Server* provides many categories of services to terminals, its data buffering service prevents transmitters retransfer lost data during network congestion or disconnection and its data transcoding service helps heterogeneous terminals communicate with each other seamlessly and efficiently.

The rest of this paper is organized as follows: Section II introduces some related works; EEWMT system architecture will be further discussed in Section III; in Section IV a videoconference through EEWMT system will be presented; conclusions and future work will be discussed in Section V.

II. RELATED WORK

WMSTFP [2] is an end-to-end TCP-friendly multimedia streaming protocol, to detect the status of the wired and wireless part of the wireless Internet, where only the last hop is wireless link as in EEWMT system. By accurately distinguishing the packet losses due to transmission errors from the congestive losses and smoothing out the pathologic round-trip-time values caused by the highly dynamic wireless environment, in WMSTFP higher throughput in wireless Internet can be achieved and transmission rate can be adjusted in a smooth and TCP-friendly manner.

The UPnP™ Forum [3] is an industry initiative designed to enable simple and robust connectivity among stand-alone devices and PCs from different vendors. The forum members are engaged in producing standards to describe device specifications usually in XML format. Our design of device profile is somehow referred to it.

iMobile [4] is a proxy-based mobile service platform designed to provide personalized services. iMobile provides a modular architecture that supports accesses from various mobile devices to various information spaces. However iMobile does not support communications between devices.

Transcoding service plays a very important roll in the design of EEWMT system. It is not a new topic, and many works have be done on it [5][6][7]. Authors of [5] proposed a video transcoding technology using intermediate data processor to enhance the quality of transcoded data. Authors of [6] proposed a video transcoding proxy for 3G wireless mobile Internet access and a video transcoding gateway for wireless access is proposed in [7]. [6] and [7] were referred to when we developed our EEWMT system, though the two are not designed to transcode end-to-end data flows.

A lot of other related works have been engaged in wireless access of multimedia data, but most of them just consider data transmissions between wireless terminals and servers.

III. SYSTEM ARCHITECTURE

EEWMT system is designed to support and provide services to end-to-end multimedia transmission, including both real time and non-real time transmission. Through EEWMT system, heterogeneous wireless devices can communicate with each other and access services provided by the system. As mentioned before, the system could be divided into three parts, *anyDevice*, *Agent Server* and *Center Server*, shown in Fig. 1.

A. *anyDevice*

anyDevice is a client-side protocol stack designed to provide services to the upper-layer protocols and applications. It collects device profile and sends it to the server, converts user commands and application signaling into formats that can be read by servers and also translates server response for users and applications. Besides, *anyDevice* is also responsible for communication control, data transmission between terminals and servers, and user profile management.

1) *User profile*

In fact, user profile is used widely in many other systems such as MSN and ICQ to record user's personal information and preferences etc., so as EEWMT system. User profile is maintained by and stored in the center server and can be accessed and edited by owners through *anyDevice*. Instead of creating a new kind of description language, we propose the usage of eXtensible Markup Language (XML) for representing user profile, as well as device profile. An example of user profile is presented as follows.

```
<?xml version="1.0"?>
<user profile>
  <id>shrek</id>
  <unique user NO>123456787654321</unique user NO>
  <name>Harry Potter</name>
  <password>*****</password>
  <email>harry.potter@hogwarts.ac.uk</email>
  ...
  <friend group>
    <name>Hogwarts School </name>
    <description>Schoolmates</description>
    ...
  </friend group>
  <friend>
    <name>Ron Weasley</name>
    <uuno>876543212345678</uuno>
    <group name>Hogwarts School</group name>
    ...
  </friend>
  ...
</user profile>
```

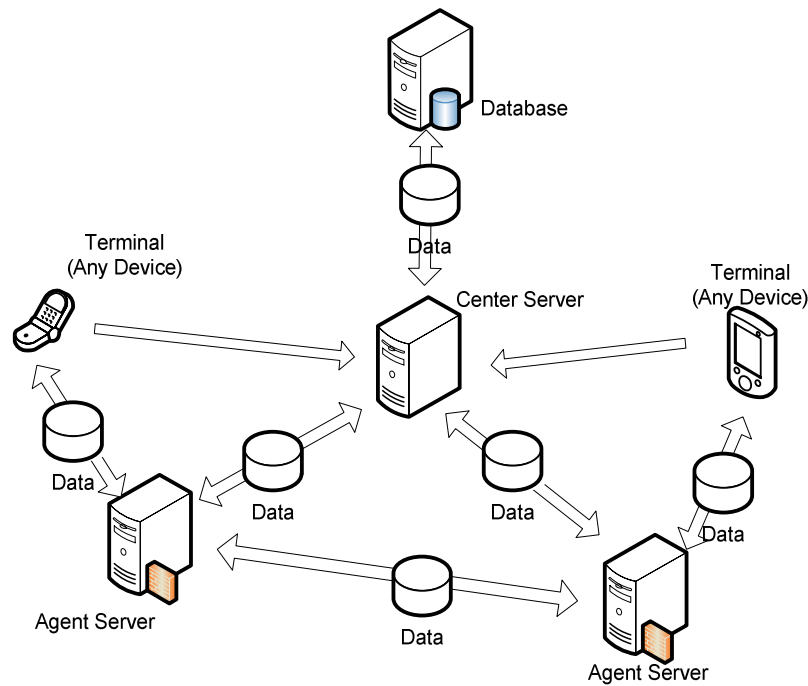


Figure 1. Framework of EEWMT system

2) Device profile

Device profile is used to describe the technical specifications and capabilities of the device such as multimedia processing capability and network transmission capability. Some other device information such as manufacturer name and device model is also presented in the profile. Device profile must be in a universal format so that servers are able to recognize all kinds of devices and provide appropriate services to them. As user profile, device profile is also given as an XML formatted file listing device specifications and capabilities, and a simple example is presented below

```
<?xml version="1.0"?>
<device information>
<deviceType>device:PDA:1</deviceType>
<manufacturer>manufacturer name</manufacturer>
<modelName>model name</modelName>
...
</device information>
<capability list>
<capability>
<name>resolution</name>
<properties>
<name>width</name>
<value>horizontal pixels</value>
<name>height</name>
<value>vertical pixels</value>
<name>depth</name>
<value>color depth</value>
<properties>
</capability>
...
</capability List>
...
</device>
```

3) Communication control

Communication control covers session initialization, sending device profile, exchange of control and indication information. It enables terminals to communicate with servers.

B. Center Server

Center Server is designed to provide indexing and database service. All terminals must connect to the center server to access the services and then send their device profile to it. The center server delivers information and data of the terminal to the agent server with the most available resources. Then a session between the agent server and the terminal is established, and the session information is also kept in center server. Whenever the session changes, the corresponding information in the server is also updated. As mentioned before, wireless connection is very vulnerable and terminals may be disconnected unexpectedly, moreover, inefficiency due to frequent disconnection and reconnection is serious. However, in EEWMT system, once a terminal loses its connection, session information kept in the center server could help the terminal renew the session quickly when it reconnects to the server, so that terminals could easily reconnect to others according to the session information, and communication efficiency is improved.

C. Agent Server

Agent servers cluster around the center server, and provide data processing and transmission services to wireless terminals, such as multimedia transcoding and data buffering services (to be detailed in following subsections). In EEWMT system, all data flows in communications must go through agent servers to be then transferred to receiving terminals. Thus, data transmissions between terminals are actually made up of transmissions between terminals and agent servers and transmissions between agent servers as illustrated in Fig. 2. If

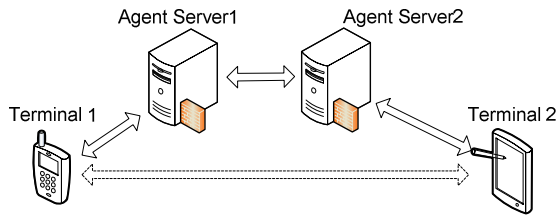


Figure 2. Communications between terminals and agent servers

agent servers to which the terminals are connecting are the same one, the transmission between servers could be skipped.

In such a design, agent servers could provide transcoding and data buffering services. This brings three advantages: easier for heterogeneous communication, lower terminal resource cost and faster recovery from connection lost.

1) Easier for heterogeneous communication

Due to the diversity of wireless terminals, data formats supported by different terminals may be not compatible with each other. It is possible that pictures or videos sent by the transmitter can't be displayed on the receiver. In such cases, not only the communication fails but the network traffic used by the transferred data is wasted. However, in EEWMT system, data flows go through agent servers, and agent servers could transcode incompatible data formats into acceptable formats according to the device profiles of the communicating terminals. Thus terminals can send and receive data in preferred formats without concerning incompatibility problems.

2) Low cost of terminal resource

In EEWMT system, terminals they need not worry about conversion of data formats, retransmission for lost packages, and reconnection to lost jobs etc., since agent servers take all these for them. This saves resources for terminals and makes them work more effectively and efficiently.

3) Faster recovery from disconnection

Connection loss is quite often for wireless connection, due to signal fading, interference and blackout etc. Once the connection between the receiver and its agent server is lost, the transmitter has to wait until it reconnects and retransmits lost packages due to disconnection. Reconnection between two wireless terminals spends much time and data retransmission is wasteful for resource-limited terminals and low-bandwidth wireless networks. It will be especially worse if the disconnected receiver had several on-going transmissions at the same time. All these do not matter in EEWMT system since terminals just connect to the specified agent server, and data buffering in agent server could keep receiving data from the sender even if the receiver is disconnected, and when the receiver is reconnected, buffered data will be delivered to it.

IV. VIDEO CONFERENCING OVER EEWMT SYSTEM

In this section, an experiment of video conferencing over EEWMT system will be presented to illustrate how it works and show the effect of communications through EEWMT system. To establish a video conference, a terminal first sends an initiate request to the server and specifies configurations of the conference, such as the conference name, the

authentication information and the number of expected participants. The server then allocates requested resources for the conference, and other terminals could join the conference after passing the authentication. During the conference, only one participant is allowed to speak at one time and the speaker's vision and voice are transferred to all other participants. Listeners could apply for the speaker-ship, and the first applicant will be the next speaker. A conference ends automatically if all participants leave. In the experiment, three types of terminals are used: PDA, laptop computer and video phone experiment board. They have different technical specifications, different terminal capabilities as listed in Table I, and supported data format are listed in table 2.

A. Video Quality

Laptop computers are obviously more powerful and support videos streams of higher resolution and higher frame rate. When one laptop computer speaks, it sends video stream of the maximal resolution 352*288 and frame rate at 30 fps. Though the PDA and experiment board have to accept video stream of lower quality after transcoded by the agent servers, the other laptop computer is able to accept the video stream without transcoding and video quality is kept.

B. Communication Stability

During conferencing, we randomly power off the experiment board and PDA to simulate sudden disconnection caused by interference. Results of the experiment show that, if the disconnected one was not the speaker, the conference was able to go on smoothly and even not aware of the lost terminals; if the speaker is disconnected, since all session information is well maintained by the center server, the speaker is able to rejoin to the conference automatically as soon as it reconnect to the system., and the conference can be renewed quickly without extra initialization.

TABLE I. TERMINALS

Terminal Type	Processor	Operating System	Multimedia Codec		Wireless Support
			Audio	Video	
PDA	Intel(R) PXA250	PocketPC 3.0	G.723.1	H.263	IEEE 802.11b
Laptop Computer	Intel PIV 1.6GHz	WinXP	G.723.1	H.263	
Experiment Board	ARM7	Linux	AMR	H.261	

TABLE II. SUPPORTED DATA FORMATS AND NETWORK BANDWIDTH

	Video Format		Audio Format	Network
	Luminance pixels(MAX)	FrameRate (MAX)fps	BitRate (MAX) kbps	Bandwidth (MAX) bps
PDA	128*96	9	20	11M
Laptop Computer	352*288	30	20	
Experiment Board	96*64	5	12.2	

C. Power saving

Power saving is very important in usage of wireless devices, especially of mobile devices such as mobile phones. Terminals that communicate through EEWMT system are believed to save more power than those communicating directly with each other. To prove that, we implement a simple videoconferencing tool for the PDA to communicate with the laptop computer directly, to be compared with videoconferencing through EEWMT system. The laptop computer sends video stream of the maximal resolution, and the PDA has to convert it to the preferred format by itself. The PDA runs out of power rapidly after a few minutes of communication, more quickly than through EEWMT system.

V. CONCLUSIONS

EEWMT system provides a solution for end-to-end wireless multimedia transmission, focused on resolving the problems due to the characteristics of wireless terminals and networks. Experiments show that, through EEWMT system, wireless devices can access more multimedia resources, communications between terminals can be achieved more easily, the communication efficiency and content quality are improved, and the workload of the terminals is reduced.

In the implementation of the system, there are still some problems that need more work to be done, such as how to define a universal device profile compatible with more devices as possible, how to build up an effective and efficient transcoding module, how to select an appropriate agent server

for the terminal, and how to coordinate the sessions between terminals and servers. Our future work will focus on resolving such problems to improve the performance of our system.

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